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WEEK-12

User-defined Functions & Recursive Functions

Ex.No. 55

Date 18.12.24

Find the Factor

Problem Statement:

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the pth element of the list, sorted ascending. If there is no pth element, return 0.

Example

$n = 20$

$p = 3$

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if $p = 3$, then 4 is returned. If $p > 6$, 0 would be returned.

Function Description

Complete the function `pthFactor` in the editor below.

`pthFactor` has the following parameter(s):

int n : the integer whose factors are to be found

int p : the index of the factor to be returned

Returns:

int: the long integer value of the pth integer factor of n or, if there is no factor at that index, then 0 is returned

Constraints

$1 \leq n \leq 1015$

$1 \leq p \leq 109$

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n , the number to factor.

The second line contains an integer p , the 1-based index of the factor to return.

Sample Input

STDIN

10

3

→

→

Function

$p = 3$

Sample Output

5

Explanation

Factoring $n = 10$ results in {1, 2, 5, 10}. Return the $p = 3$ rd factor, 5, as the answer.

Program:

```
1  /*
2   * Complete the 'pthFactor' function below.
3   *
4   * The function is expected to return a LONG_INTEGER.
5   * The function accepts following parameters:
6   * 1. LONG_INTEGER n
7   * 2. LONG_INTEGER p
8   */
9
10 long pthFactor(long n, long p)
11 {
12     int count=0;
13     for(long i=1;i<=n;i++)
14     {
15         if(n%i==0)
16         {
17             count++;
18             if(count==p)
19             {
20                 return i;
21             }
22         }
23     }
24     return 0;
25 }
```

	Test	Expected	Got	
✓	<code>printf("%ld", pthFactor(10, 3))</code>	5	5	✓
✓	<code>printf("%ld", pthFactor(10, 5))</code>	0	0	✓
✓	<code>printf("%ld", pthFactor(1, 1))</code>	1	1	✓

Passed all tests! ✓

Ex.No. 56

Date 18.12.24

4th Bit

Problem Statement:

A binary number is a combination of 1s and 0s. Its nth least significant digit is the nth digit starting from the right starting with 1. Given a decimal number, convert it to binary and determine the value of the the 4th least significant digit.

Example

number = 23

Convert the decimal number 23 to binary number: $23_{10} = 2^4 + 2^2 + 2^1 + 2^0 = (10111)_2$.

The value of the 4th index from the right in the binary representation is 0.

Function Description

Complete the function fourthBit in the editor below.

fourthBit has the following parameter(s):

int number: a decimal integer

Returns:

int: an integer 0 or 1 matching the 4th least significant digit in the binary representation of number.

Constraints

$0 \leq \text{number} < 231$

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.
The only line contains an integer, number.

Sample Input

STDIN	Function
32	fourthBit = 32
-----	-----
→	

Sample Output

0

Explanation

Convert the decimal number 32 to binary number: $32_{10} = (100000)_2$.

The value of the 4th index from the right in the binary representation is 0.

Program:

```

1  /*
2   * Complete the 'fourthBit' function below.
3   *
4   * The function is expected to return an INTEGER.
5   * The function accepts INTEGER number as parameter.
6   */
7
8  int fourthBit(int number)
9  {
10     int binary[32];
11     int i=0;
12     while(number>0)
13     {
14         binary[i]=number%2;
15         number/=2;
16         i++;
17     }
18     if(i>=4)
19     {
20         return binary[3];
21     }
22     else
23     return 0;
24 }
25

```

	Test	Expected	Got	
✓	printf("%d", fourthBit(32))	0	0	✓
✓	printf("%d", fourthBit(77))	1	1	✓

Passed all tests! ✓

Ex.No. 57

Date 18.12.24

The Power Sum

Problem Statement:

Find the number of ways that a given integer, X , can be expressed as the sum of the N th powers of unique, natural numbers.
For example, if $X = 13$ and $N = 2$, we have to find all combinations of unique squares adding up to 13. The only solution is $2^2 + 3^2$.

Function Description

Complete the powerSum function in the editor below. It should return an integer that represents the number of possible combinations.

powerSum has the following parameter(s):

X : the integer to sum to

N : the integer power to raise numbers to

Input Format

The first line contains an integer X .

The second line contains an integer N .

Constraints

$1 \leq X \leq 1000$

$2 \leq N \leq 10$

Output Format

Output a single integer, the number of possible combinations calculated.

Sample Input

10

2

Sample Output

1

Explanation

If $X = 10$ and $N = 2$, we need to find the number of ways that 10 can be represented as the sum of squares of unique numbers.

$10 = 1^2 + 3^2$

This is the only way in which 10 can be expressed as the sum of unique squares.

Program:

```

1  /*
2   * Complete the 'powerSum' function below.
3   *
4   * The function is expected to return an INTEGER.
5   * The function accepts following parameters:
6   * 1. INTEGER x
7   * 2. INTEGER n
8   */
9
10 int powerSum(int x, int m, int n)
11 {
12     int power =m;
13     for(int i=1;i<n;i++){
14         power*=m;
15     }
16     if(power>x){
17         return 0;
18     }
19     if(power == x){
20         return 1;
21     }
22     return powerSum(x-power,m+1,n) + powerSum(x,m+1,n);
23 }

```

	Test	Expected	Got	
✓	printf("%d", powerSum(10, 1, 2))	1	1	✓

Passed all tests! ✓

Ex.No. 58

Date 18.11.24

Hack the Money

Problem Statement:

You are a bank account hacker. Initially you have 1 rupee in your account, and you want exactly N rupees in your account. You wrote two hacks, first hack can multiply the amount of money you own by 10, while the second can multiply it by 20. These hacks can be used any number of time. Can you achieve the desired amount N using these hacks.

Constraints:

$$1 \leq T \leq 100$$

$$1 \leq N \leq 10^{12}$$

Input

The test case contains a single integer N.

Output

For each test case, print a single line containing the string "1" if you can make exactly N rupees or "0" otherwise.

SAMPLE INPUT

1

SAMPLE OUTPUT

1

SAMPLE INPUT

2

SAMPLE OUTPUT

0

Program:

```

1  /*
2   * Complete the 'myFunc' function below.
3   *
4   * The function is expected to return an INTEGER.
5   * The function accepts INTEGER n as parameter.
6   */
7
8  int myFunc(int n)
9  {
10     return n%10==0 || n==1;
11 }
12

```

	Test	Expected	Got	
✓	printf("%d", myFunc(1))	1	1	✓
✓	printf("%d", myFunc(2))	0	0	✓
✓	printf("%d", myFunc(10))	1	1	✓
✓	printf("%d", myFunc(25))	0	0	✓
✓	printf("%d", myFunc(200))	1	1	✓

Passed all tests! ✓